

TimeCoins: A Digital Currency Backed by Shares

A PROJECT REPORT

for

Major Project-1 (CA301P)

Session (2025-26)

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**Submitted in partial fulfilment of the
Requirements for the Degree of**

MASTER OF COMPUTER APPLICATION

**Under the Supervision of
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Submitted to

**DEPARTMENT OF COMPUTER APPLICATIONS
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(NOVEMBER- 2025)

CERTIFICATE

Certified that **Aman Kumar 202410116100020, Deepu Kumari 202410116100057, Anand Patel 202410116100023** has/ have carried out the project work having **“TimeCoins: A Digital Currency Backed by Shares” (Major Project-I, CA301P)** for **Master of Computer Application** from Dr. A.P.J. Abdul Kalam Technical University (AKTU) (formerly UPTU), Lucknow under my supervision. The project report embodies original work, and studies are carried out by the student himself/herself and the contents of the project report do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.

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ABSTRACT

The rapid evolution of digital financial systems has led to the development of innovative currency models that integrate real-world market value with secure online transactions. **TimeCoins** is a share-backed digital currency designed to provide transparent, secure, and scalable financial interactions. In this system, the value of TimeCoins is directly influenced by the share market, ensuring real asset-based valuation rather than speculative inflation.

A unique feature of TimeCoins is its **community-driven coin generation model**. When a new company proposes investment by offering a percentage of its shares, it is listed on the platform for user voting. Based on user approval and analytical evaluation, the company may be accepted, enabling the generation of new TimeCoins proportional to the contributed share value. This approach promotes user participation, fair decision-making, and responsible currency growth.

To support seamless user communication, the system includes a **hybrid real-time chat module**, where messages are sent via HTTP requests and delivered to recipients instantly using WebSocket technology. The platform is developed using **Spring Boot** for the backend, **React with Vite** for the frontend, and **MySQL** for data storage. **JWT-based authentication** is implemented to ensure secure access and protect financial operations.

Overall, TimeCoins integrates secure transactions, real-time interaction, and community governance to build a modern digital currency ecosystem backed by trusted financial assets.

Keywords: Digital Currency, Share-Backed Asset, Community-Driven Coin Generation, Secure Transactions.

ACKNOWLEDGEMENTS

Success in life is never attained single-handedly. My deepest gratitude goes to my project supervisor, **Ms. Hunny Gaur**, for her guidance, help, and encouragement throughout my project work. Their enlightening ideas, comments, and suggestions.

Words are not enough to express my gratitude to **Dr. Sachin Malhotra**, Professor and Dean, Department of Computer Applications, for his insightful comments and administrative help on various occasions.

Fortunately, I have many understanding friends, who have helped me a lot on many critical conditions.

Finally, my sincere thanks go to my family members and all those who have directly and indirectly provided me with moral support and other kind of help. Without their support, completion of this work would not have been possible in time. They keep my life filled with enjoyment and happiness.

AMAN KUMAR

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CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

In today's digital era, the financial sector is rapidly adopting technology-driven solutions to enhance the security, transparency, and efficiency of transactions. Digital currencies have emerged as a powerful alternative to traditional financial systems. **TimeCoins** is a share-backed digital currency designed to link virtual currency value with real-world share market assets. The platform enables secure financial transactions, real-time communication, and user-driven currency expansion. By integrating voting-based approval of new share investments, TimeCoins introduces a democratic and trustworthy model for new coin generation.

1.2 SYSTEM OBJECTIVES

The major objectives of the TimeCoins system are:

- To create a digital asset whose value is derived from actual market-backed shares.
- To implement a **community governance model** for approving new investments.
- To provide secure and authenticated user access using **JWT authentication**.
- To enable **real-time messaging** and transaction updates through WebSocket technology.
- To maintain transparency in coin distribution and share contribution.
- To build a scalable and user-friendly financial interaction platform.

1.3 FUNCTIONALITY

TimeCoins offers the following key functionalities:

- **Share-based Coin Generation:** Approval of new companies through user voting followed by coin creation based on contributed share value.
- **Secure User Authentication** using JWT for login and financial operations.
- **Real-time Chat System** using hybrid messaging (HTTP + WebSocket).
- **Transaction Management** including sending, receiving, and storing transaction history.
- **Email Notification System** for voting participation and updates.

- **User Dashboard** for accessing wallet, voting panel, and communication features.

1.4 SIGNIFICANCE

TimeCoins provides a practical innovation in digital finance through:

- **Economic Stability:** Coin value depends on real share market performance.
- **User Trust & Transparency:** Community voting ensures fair decision-making.
- **Increased Security:** JWT access control and verified transactions protect users.
- **Open Communication:** Real-time messaging improves engagement within the ecosystem.
- **Future Scalability:** New companies and users can join, increasing market strength.

1.5 DATA COMMUNICATION

Communication in TimeCoins occurs in two forms:

- **HTTP REST API** for sending user requests such as login, voting, and chat message creation.
- **WebSocket Channel** for delivering live updates such as instant message notifications and transaction status.

This hybrid model ensures both data reliability and real-time interaction.

1.6 BASIC COMMUNICATION MODEL

The system employs a **client-server communication model**. Users (clients) access the platform through a web application built with React and Vite. The backend server built with Spring Boot handles business logic, authentication, messaging, coin generation, and database operations. MySQL stores persistent data, while WebSockets provide continuous two-way communication for instant event delivery. This model supports high responsiveness and secure data exchange.

MySQL databases store and retrieve user data securely and efficiently.

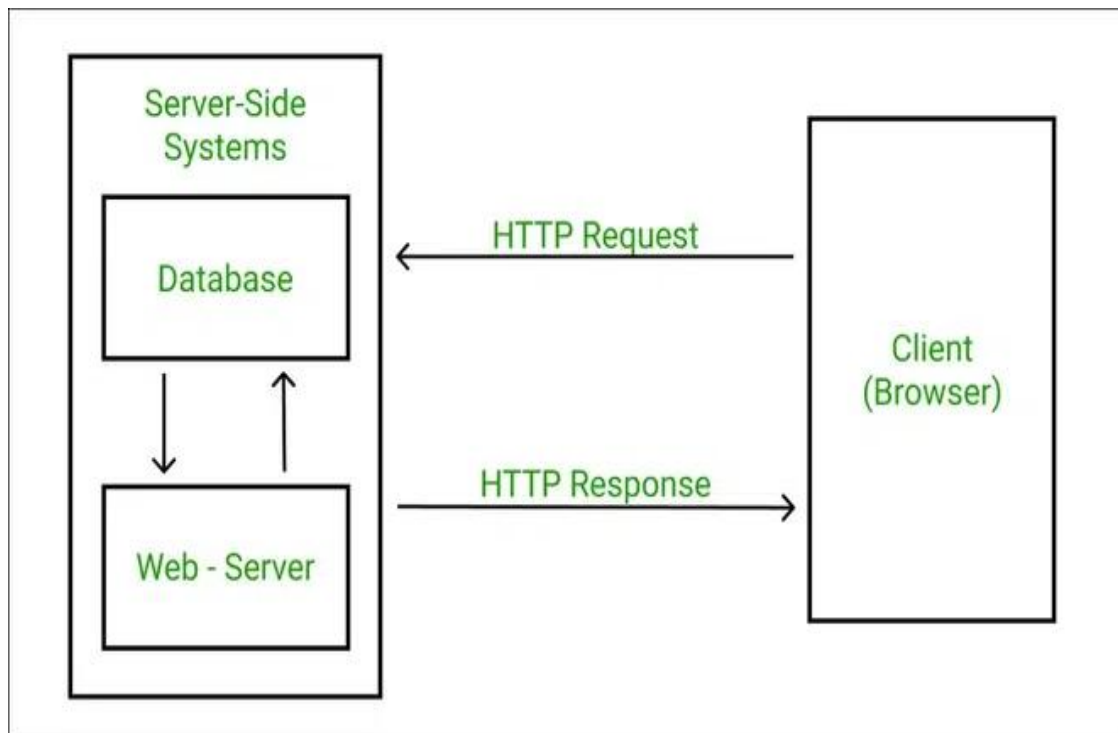


Figure 1.1 Database Architecture

1.7 REPORT STRUCTURE

This report is arranged into the following chapters:

- **Introduction** – Overview, significance, objectives, and communication model.
- **Literature Review** – Study of existing digital currency systems and related technologies.
- **System Analysis** – Requirements, existing vs. proposed system comparison.
- **System Design** – Architecture, database schema, module diagrams, and workflows.
- **Implementation** – Technologies used, coding strategies, and module development.
- **Testing** – Test cases, results, and validation of system performance.
- **Conclusion and Future Scope** – Achievements, limitations, and potential enhancements.

CHAPTER 2

Feasibility Study

2 Feasibility Study

A feasibility study is conducted to evaluate whether the proposed system is practical, technically valid, and economically beneficial. For the TimeCoins project, three major feasibility factors are analyzed: **Technical**, **Operational**, and **Economic** feasibility.

2.1 Technical Feasibility

The TimeCoins platform is developed using established and widely supported technologies including **Spring Boot**, **React with Vite**, and **MySQL**. These technologies provide:

- **Strong Backend Performance** through Spring Boot framework with secure and scalable APIs.
- **Efficient Frontend Development** using React (Vite) for faster UI rendering and modular components.
- **Real-Time Communication** via WebSocket for hybrid messaging and live notifications.
- **Secure Authentication** using JWT for controlled access and protection against unauthorized activities.
- **Reliable Data Management** using MySQL relational database for structured data storage.

All required tools, frameworks, and hardware resources are easily available, ensuring seamless development and deployment. Therefore, the project is **technically feasible**.

2.2 Operational Feasibility

The system is designed to offer an intuitive and user-friendly interface, ensuring that users can easily:

- Register and authenticate securely.
- Perform TimeCoin transactions.
- Vote for company approvals.
- Access wallet, balance history, and notifications.

- Communicate via live chat in real time.

Administrators can efficiently manage company proposals, analyze voting results, and oversee user activities. The strong integration of notifications and real-time updates enhances user engagement and reliability. Proper training and documentation make the system simple to operate. Thus, TimeCoins is **operationally feasible**.

2.3 Economic Feasibility

TimeCoins utilizes:

- **Open-source technologies** → no high-cost licensing
- Standard hardware and networking infrastructure
- Scalable architecture to reduce future cost overhead

The system can potentially generate revenue in the future through:

- Transaction/service fees
- Partnerships with approved companies
- Increased market adoption of TimeCoins

The long-term benefits exceed the initial development and maintenance expenses. Therefore, the project is **economically feasible** and capable of providing profitable outcomes.

CHAPTER 3

Literature Review

3. Literature Review

The increasing popularity of digital currencies and real-time financial communication systems has motivated researchers and organizations to innovate secure and value-driven digital ecosystems. This chapter reviews the existing technologies, models, and systems that have influenced the development of TimeCoins.

3.1 Digital Currency Systems

Digital currencies such as **Bitcoin** and **Ethereum** operate on decentralized blockchain technology. Their market value is primarily driven by **speculation and demand**, with no direct link to real-world assets. Although secure and transparent, such currencies experience high volatility, making them less suitable for stable financial transactions.

TimeCoins improves this concept by **backing currency value with actual company shares**, providing a more reliable and economically grounded valuation model.

3.2. Share-Backed Financial Models

Traditional stock markets allow users to invest in company shares, but these systems are:

- Often complex for general users
- Centralized and fully controlled by governing bodies
- Slow in transaction processing

TimeCoins combines digital currency flexibility with **share market backing**, offering:

- A democratic decision-making process through **user voting**
- A straightforward method for investment-driven coin creation
- Better transparency and public participation in financial growth

3.3. Community-Driven Governance Models

Some platforms enable **Decentralized Autonomous Organizations (DAOs)** where users vote on decisions related to currency or platform growth. However, most DAOs focus only on policy control rather than **asset-backed coin generation**.

TimeCoins leverages community approval **as a core mechanism** to expand currency supply responsibly.

3.4. Real-Time Communication Technologies

Modern applications use WebSocket technology for:

- Instant updates
- Live messaging
- Faster event-driven data exchange

Hybrid communication systems combining **HTTP REST APIs with WebSockets** are widely used in platforms such as social networks, gaming systems, and trading apps.

TimeCoins adopts this model to support:

- Real-time messaging
- Instant transaction status updates
- Notification delivery during voting phases

3.5. Secure Authentication and Data Management

JWT (JSON Web Token) is widely used for:

- Access control
- Session management
- Protecting sensitive operations

MySQL provides a reliable and structured approach to financial and user data storage.

Together, these technologies ensure that TimeCoins maintains:

- Strong security layers
- Data integrity
- Efficient backend performance

CHAPTER 4

Project Objective

The primary objective of the TimeCoins system is to create a secure, transparent, and innovative digital currency ecosystem backed by real-world share market assets. TimeCoins aims to connect digital financial models with community decision-making and real-time communication, offering users a stable and highly interactive economic platform.

The key objectives of the project are as follows:

1. **To develop a digital currency backed by company shares**
 - Ensuring real asset-based valuation for financial stability.
2. **To introduce a user-governed coin generation mechanism**
 - Allowing users to vote on company proposals before coin creation.
3. **To support secure and authenticated financial operations**
 - Using JWT-based authorization for login and all sensitive actions.
4. **To provide a real-time communication system between users**
 - Through hybrid messaging using HTTP and WebSocket technology.
5. **To enhance transparency in digital economic growth**
 - Showing how share contributions influence coin supply and value.
6. **To enable future e-commerce integration**
 - Allowing users to purchase products using TimeCoins, indirectly paying through share value.
7. **To improve user engagement in financial decision-making**
 - Notifications and voting participation improve trust and activity.
8. **To build a scalable and secure digital finance platform**
 - Capable of supporting a growing number of users and company partnerships.

Long-Term Vision

- To establish TimeCoins as a **next-generation asset-based payment system**
- To revolutionize online shopping by allowing **share-backed purchases**
- To promote fair and community-driven economic expansion

CHAPTER 5

Hardware and Software Requirements

A robust and efficient system design requires suitable hardware and software resources to ensure smooth development, deployment, and execution. TimeCoins uses modern development tools and efficient communication technologies to provide scalability, security, and high availability.

5.1 Hardware Requirements

The minimum hardware configuration required for development and execution is as follows:

Component	Requirement
Processor	Intel Core i3 or higher
RAM	8 GB (minimum), 16 GB recommended
Hard Disk	Minimum 256 GB storage
Display	1024 × 768 resolution or higher
Network	Stable broadband connection for real-time data exchange

For server deployment, higher hardware capabilities may be required based on the number of active users.

5.2 Software Requirements

Category	Software Used	Purpose
Operating System	Windows / Linux / macOS	Development and deployment environment
Backend Framework	Spring Boot	API development & business logic
Frontend Framework	React + Vite	UI development and fast build system
Database	MySQL	Structured data storage

Category	Software Used	Purpose
Authentication	JWT (JSON Web Token)	Secure login and access control
Real-Time Technology	WebSocket	Hybrid messaging system
Programming Language	Java (Backend), JavaScript (Frontend)	Core development languages
Build Tools	Maven, npm	Dependency management
IDE/Editors	IntelliJ IDEA / Eclipse / VS Code	Code development tools
Version Control	Git, GitHub	Code collaboration & repository management
Email Service	SMTP Mail Server	Voting and notification alerts

Additional Tools and Libraries

- Redux Toolkit → Global state management in frontend
- Axios → API communication
- Spring Security → Authentication and authorization
- Node.js → Frontend package management and builds
- WebSocket Client Libraries → Real-time communication support

5.3 Development Environment

The TimeCoins platform was developed in a **full-stack environment** using both frontend and backend technologies to ensure smooth system integration and data consistency.

The frontend is created using **React with Redux Toolkit**, which manages application state and authentication.

The screenshot below shows the authSlice.js file implemented in Visual Studio Code, where the authentication logic is defined for login and token storage.

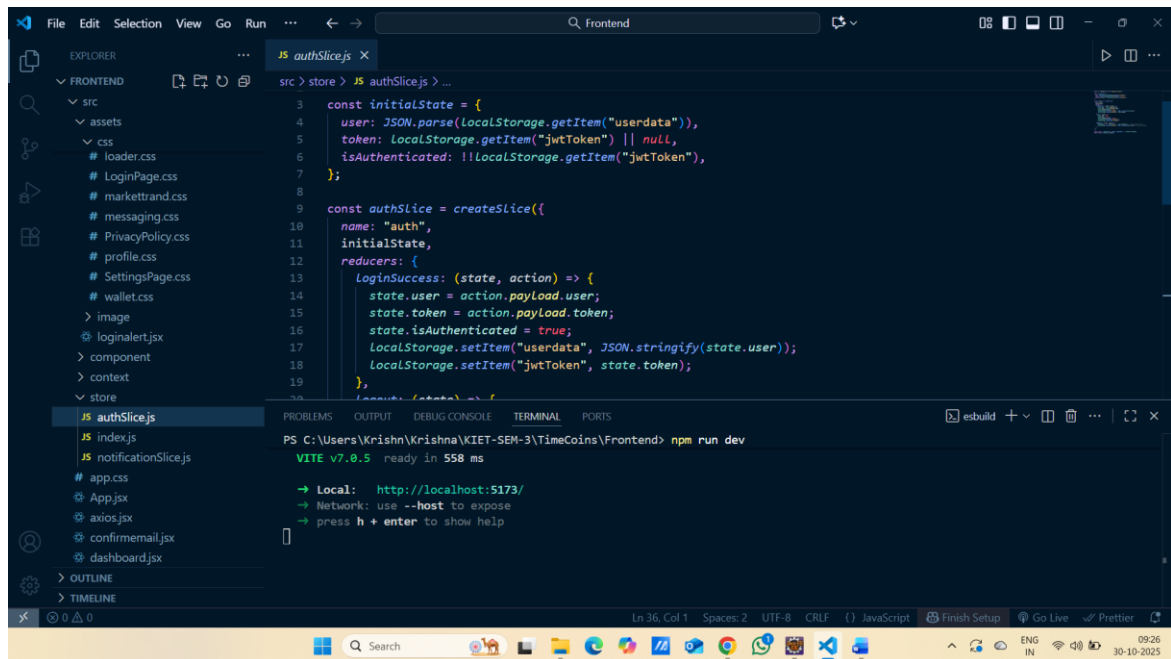


Figure 5.1 Frontend development in Visual Studio Code showing Authentication Slice implementation.

The backend is built using Spring Boot within Eclipse IDE, providing secure RESTful APIs for data exchange between users, companies, and the database.

The image below shows the FluctuationController.java class, which handles endpoints such as /api/value and /company/add, responsible for managing coin fluctuations and company operations.

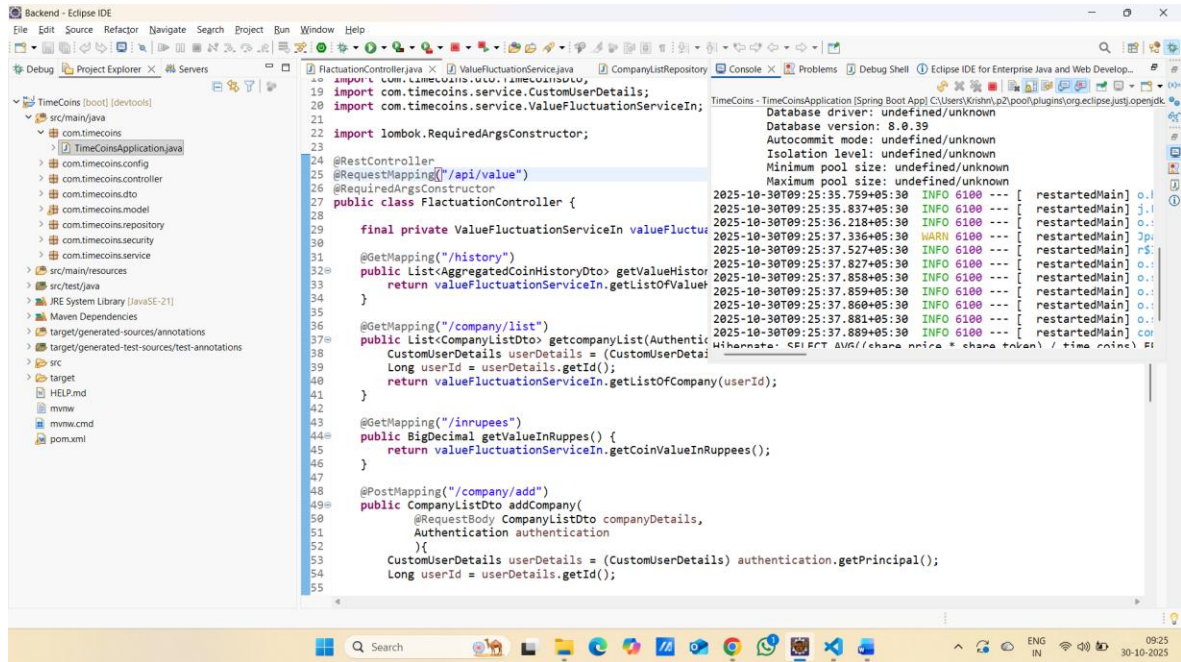


Figure 5.2: Backend development in Eclipse IDE showing REST API Controller implementation.

CHAPTER 6

Project Flow

6.1 Requirement Gathering and Analysis

- I identified the primary functional requirements such as user registration, login, wallet creation, TimeCoins transfer, transaction history, and product purchasing via TimeCoins.
- Studied security requirements including password encryption, role-based authentication, and secure transactions.
- Understood business rules: limited supply of TimeCoins, share-based valuation, and demand-driven price fluctuations.
- Prepared SRS (Software Requirement Specification) to finalize application scope.

6.2 System Design

Designed the system architecture following a secure and scalable layered structure:

- **Presentation Layer** → UI handling user interactions
- **Business Layer** → Controller & Service handling logic
- **Data Layer** → Repository & Database storage

Focus on modular design for easier enhancements, wallet APIs, messaging services, and integrations.

6.3 Technology Selection

- **Backend:** Spring Boot (Java), REST API
- **Frontend:** React with VITE
- **Database:** MySQL
- **Security:** Spring Security + JWT Authentication
- **Others:** WebSocket (Real-time transactions & messages)

Reason for selection:

High scalability, enterprise-level security, efficient communication between components, and strong community support.

6.4 Development

- Implemented REST APIs for complete wallet lifecycle:
 - User Registration, Login, Logout

- Wallet Initialization
- Secure Payment Authorization
- Transaction History Logging (Success & Failure)
- Introduced Business Rules:
 - TimeCoins mapped to company share value
 - Value updated dynamically based on demand
- Implemented WebSocket notifications for real-time transaction status.
- Validations added for incorrect password, insufficient balance, and unauthorized access.

6.5 Deployment

- Application packaged using Maven build.
- Deployed on cloud environment / local server using:
 - Apache Tomcat
 - MySQL Cloud/Local Database
- Continuous testing & monitoring enabled before production release.

6.7 Maintenance and Support

- Continuous monitoring of API performance and security vulnerabilities.
- Bug fixes and updates based on user feedback.
- Backup mechanisms and logs maintained for all financial actions.

6.8 Future Enhancements

- Integration with major e-commerce companies (Like Flipkart/Amazon - users buy products using TimeCoins which represent indirect shares).
- Mobile application version for Android & iOS.
- AI-powered fraud detection.
- TimeCoins trading marketplace where users buy/sell TimeCoins openly.

6.9 Use Case Diagram

A use case diagram is used to represent the dynamic behavior of a system. It encapsulates the system's functionality by incorporating use cases, actors, and their relationships. It models the tasks, services, and functions required by a

system/subsystem of an application. It depicts the high-level functionality of a system and also tells how the user handles a system. Following are the purposes of a use case diagram given below:

- It gathers the system's needs.
- It depicts the external view of the system.
- It recognizes the internal as well as external factors that influence the system.
- It represents the interaction between the actors.

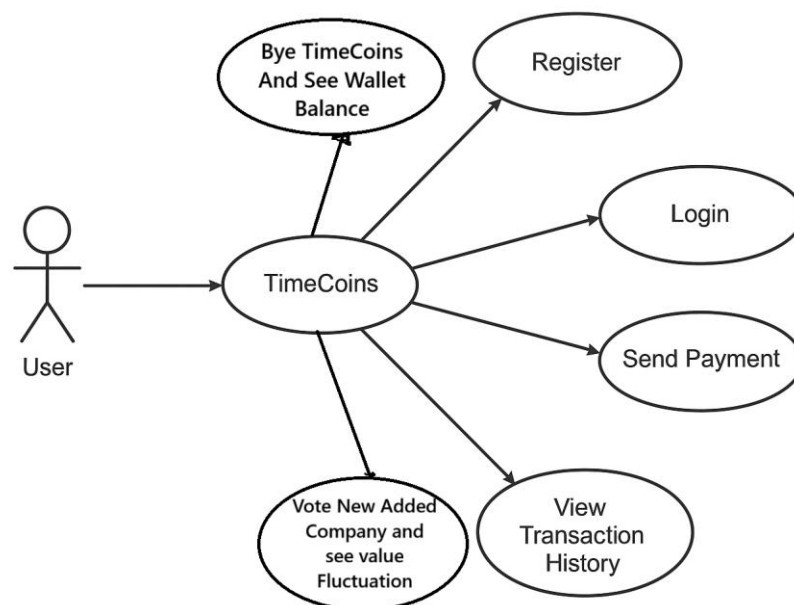


Figure 6.1 Use Case Diagram

6.10 Entity Relationship (ER) Diagram

The **Entity Relationship Diagram (ERD)** represents the logical structure of the TimeCoins database. It illustrates how various entities interact within the system, ensuring data integrity and smooth communication between modules.

Major entities include **Users**, **Companies**, **Transactions**, **Votes**, **Messages**, and **Coin Value History**.

Each entity stores relevant attributes such as transaction details, company information, user credentials, and voting data.

Relationships between these entities are designed using **primary and foreign keys**, maintaining referential integrity across the system.

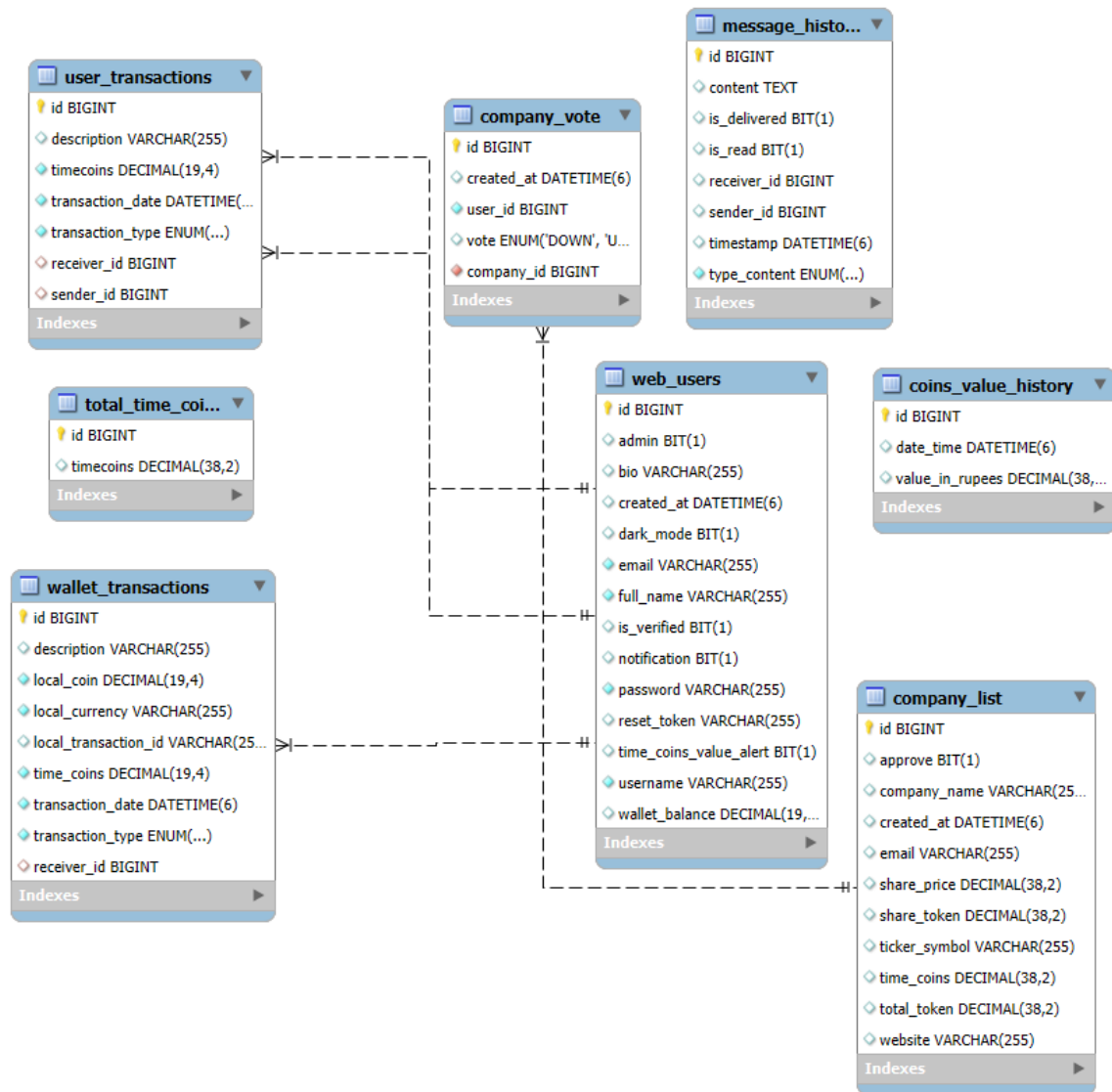


Figure 6.2: Entity Relationship Diagram (ERD) of the TimeCoins System.

- **web_users** - Stores user information, login credentials, and wallet balance.
- **company_list** - Contains details of registered companies and their share-based TimeCoins values.
- **company_vote** - Manages user voting for company approval before TimeCoin generation.
- **wallet_transactions** - Logs user-to-user transactions and purchases of TimeCoins.
- **user_transactions** - Records all types of transfers including send, receive, and purchase operations.

- **message_history** - Stores chat history between users for real-time communication.
- **coins_value_history** - Tracks coin value fluctuations based on share value changes.
- **total_time_coins** - Keeps record of total TimeCoins in circulation across the system.

The database is **normalized** to reduce redundancy and improve query performance. All relationships follow a **one-to-many** or **many-to-one** structure, ensuring scalability and reliability.

CHAPTER 7

Project Outcome

7.1 Overview of the System

The TimeCoins project has been successfully designed and implemented to provide a secure and interactive platform where users can manage digital coins that are directly mapped to company shares.

The system integrates multiple modules such as **user management, wallet transactions, company approval and voting, real-time messaging, and dynamic coin value calculation.**

This project demonstrates how digital currency can be tied to real-world assets (shares), introducing a new economic model where users indirectly trade and purchase products using value-backed coins.

7.2 System Working and Interface

This section presents the working modules and user interfaces of the TimeCoins system. Each screenshot demonstrates a core functional component of the application.

7.2.1 Login and Registration Module

This module ensures secure user authentication through encrypted credentials and JWT tokens.

New users can register with required details, while existing users can log in to access their wallets and dashboard.

**TimeCoins**
The Coin That Moves with Time

Login

Register

Forgot

Welcome Back

Login

[Forgot Password?](#)

[New User? Register](#)

**TimeCoins**
The Coin That Moves with Time

Login

Register

Forgot

Create Account

Register

[Already have an account? Login](#)

Figure 7.1: Login and Registration Page of the TimeCoins Application.

7.2.2 User Dashboard

The **TimeCoins Dashboard** serves as the central hub for users to manage all major functionalities within the platform. Once logged in, users can easily navigate through key modules such as **Wallet Balance**, **Send Coins**, **Messages**, **Transaction History**, **Coin Value**, and **Profile Management**.

Each feature is clearly represented through interactive cards, ensuring a seamless and user-friendly experience.

The dashboard also displays **User History**, which helps track recent interactions and connected accounts.

This interface demonstrates a clean and modern UI design built using **React and Vite**, ensuring fast performance and responsive behavior. The design emphasizes simplicity, enabling users to efficiently access their TimeCoins data and actions.

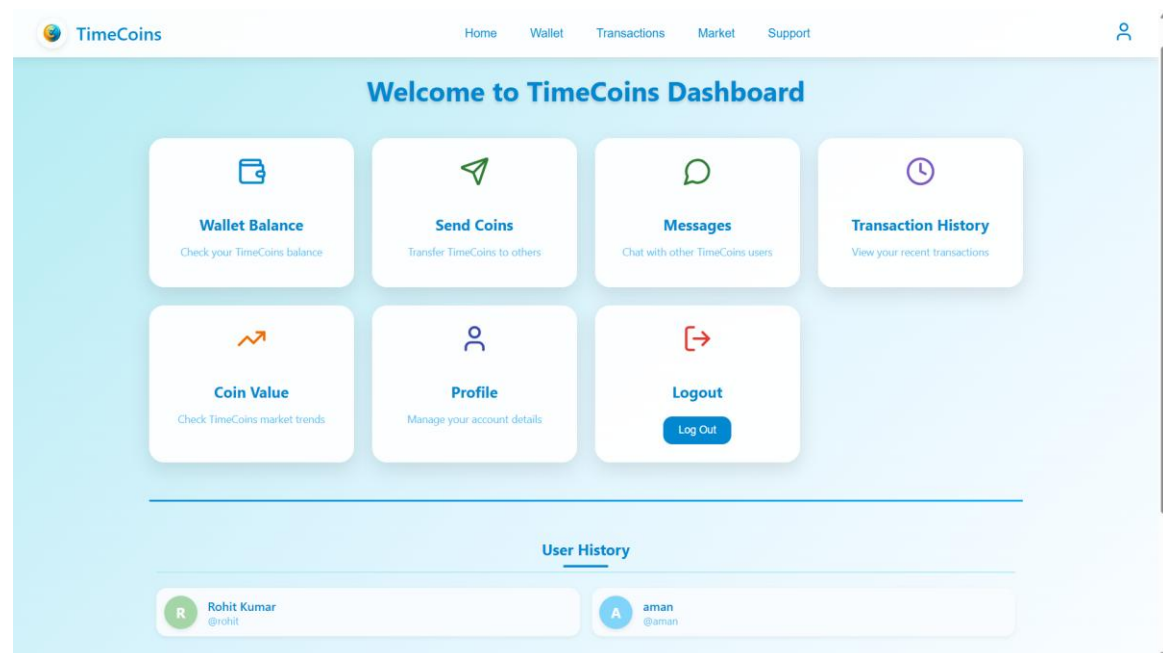


Figure 7.2: User Dashboard displaying key features like Wallet Balance, Send Coins, Messages, Transaction History, Coin Value, and Profile Management.

7.2.3 Wallet Management Interface

The **Wallet Page** in the TimeCoins system allows users to manage and monitor their digital balance efficiently. The interface clearly displays the **current TimeCoins balance**, ensuring transparency and real-time tracking of digital assets.

Users can also **add money to their wallet** through integrated payment gateways like PayPal. Upon successful transactions, detailed **transaction history** records are displayed, including the **transaction ID, deposit amount, description, payment value in USD, and PayPal Transaction ID**.

This design ensures simplicity and security, providing users with confidence while managing their TimeCoins assets. The wallet is tightly integrated with the backend via **Spring Boot APIs** to maintain synchronized and secure transaction data.

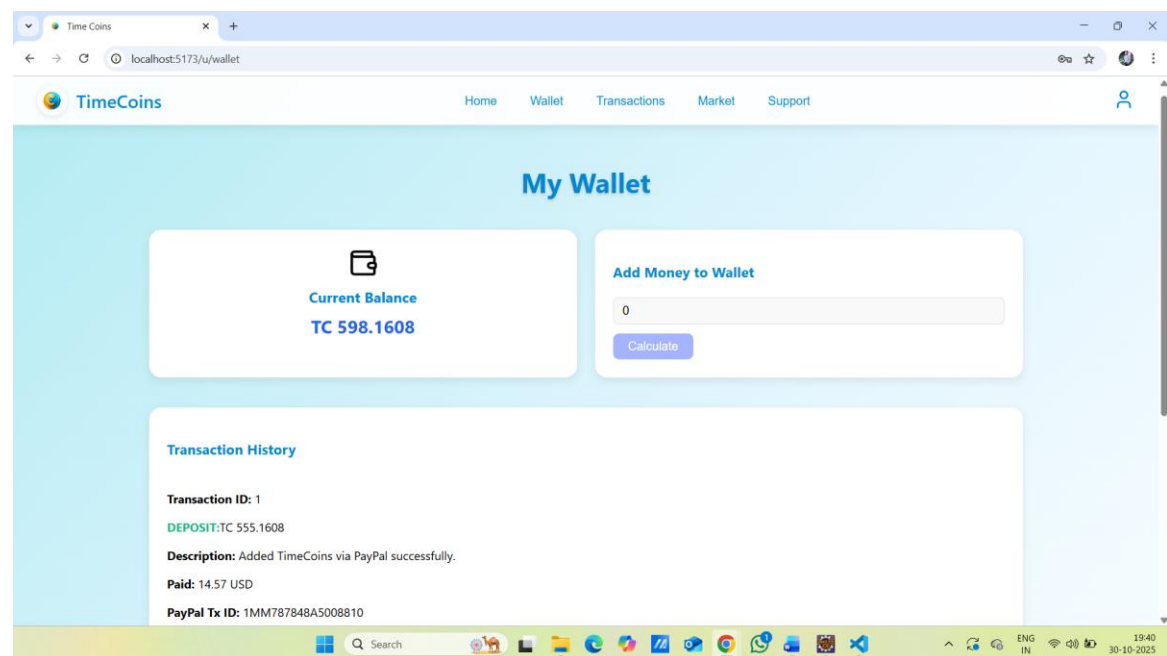


Figure 7.3: Wallet Management Interface showing current balance, add-money functionality, and detailed transaction history.

7.2.4 Real-Time Messaging and Transaction Interface

The **Messaging Page** in the TimeCoins system enables users to communicate and transact simultaneously within a single interface. This module integrates **real-time chatting** with **instant TimeCoin transfer functionality**, creating a hybrid communication model.

Messages are sent using **HTTP requests**, while live updates are handled through **WebSocket connections**, ensuring smooth and immediate message delivery.

Users can view transaction confirmations directly in the chat window—such as “Transfer Successful” or “Insufficient Balance”—providing instant feedback about their TimeCoin transactions.

This design improves user engagement and convenience by allowing peer-to-peer digital payments during active conversations, making financial interactions seamless within the communication flow.

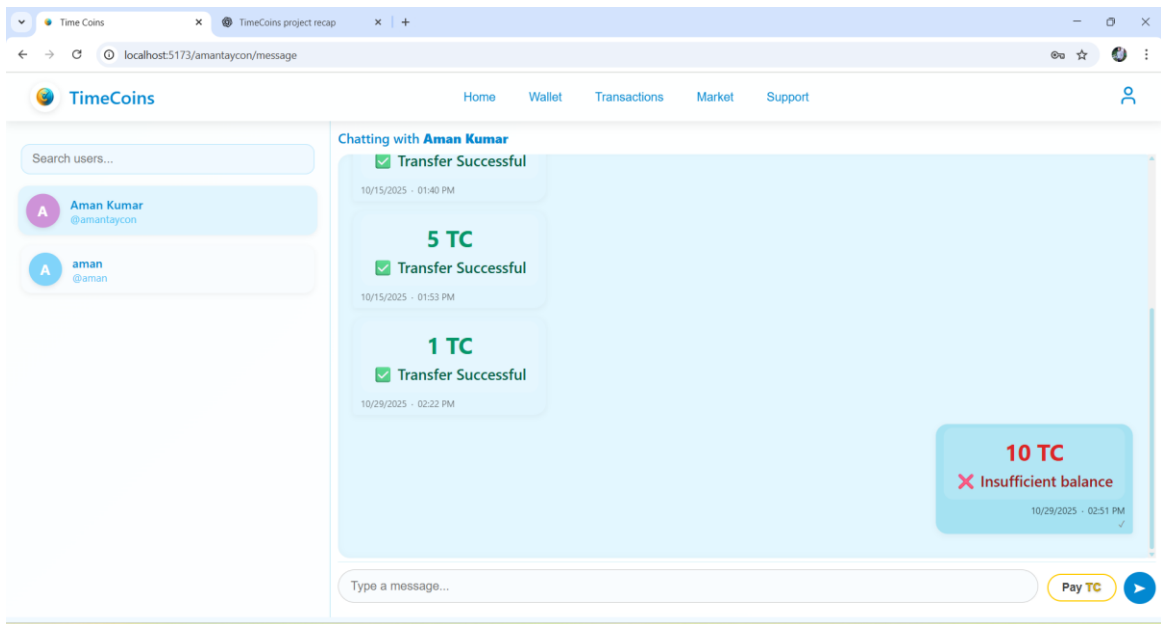


Figure 7.4: Real-time Messaging Interface showing chat-based TimeCoin transfers and transaction status notifications.

7.2.5 Transaction History Module

The **Transaction History** page provides users with a comprehensive record of all TimeCoin-related activities, ensuring full transparency in financial operations.

Each transaction entry includes crucial details such as **Transaction ID**, **Type (Credit/Debit/Failed)**, **Amount**, **Date**, **Payer**, and **Receiver**. This structured format allows users to easily review their past transactions and monitor their TimeCoin usage patterns.

The system automatically records both successful and failed transactions, helping users identify issues like insufficient balance or failed transfer attempts.

This feature enhances **financial accountability**, supports **error tracking**, and helps maintain a **secure transaction environment** for all users.

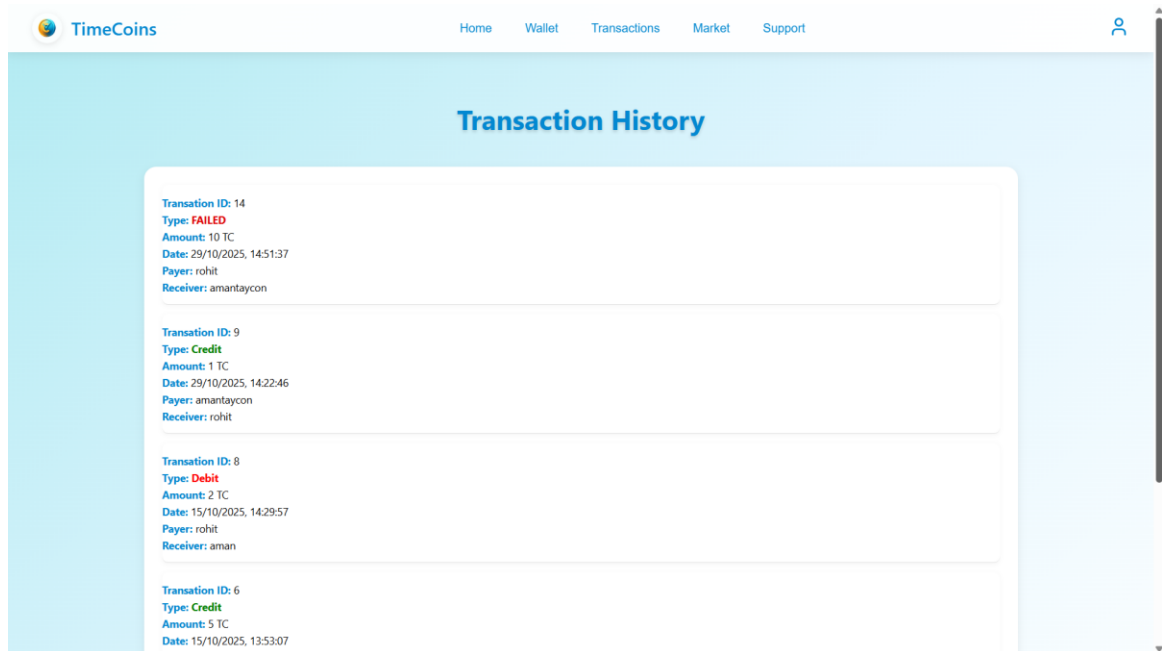


Figure 7.5: Transaction History page displaying detailed logs of all successful and failed TimeCoins transactions.

7.2.6 Market Trends and Coin Analysis

This section visualizes the TimeCoins' market value fluctuations over time. It provides a clear representation of price trends, helping users analyze how the share market and demand affect the TimeCoin's value. The dropdown filter allows users to view trends across different time periods or companies.



Figure 7.6: Market Trends and Coin Analysis

7.2.7 Coin Linked Companies Page

This section displays all companies whose shares are mapped to TimeCoins. Each company listing shows key details such as ticker name, mapped TimeCoins value, approval status, and user voting results (Upvote/Downvote). This feature helps users understand how different companies influence the overall value of TimeCoins

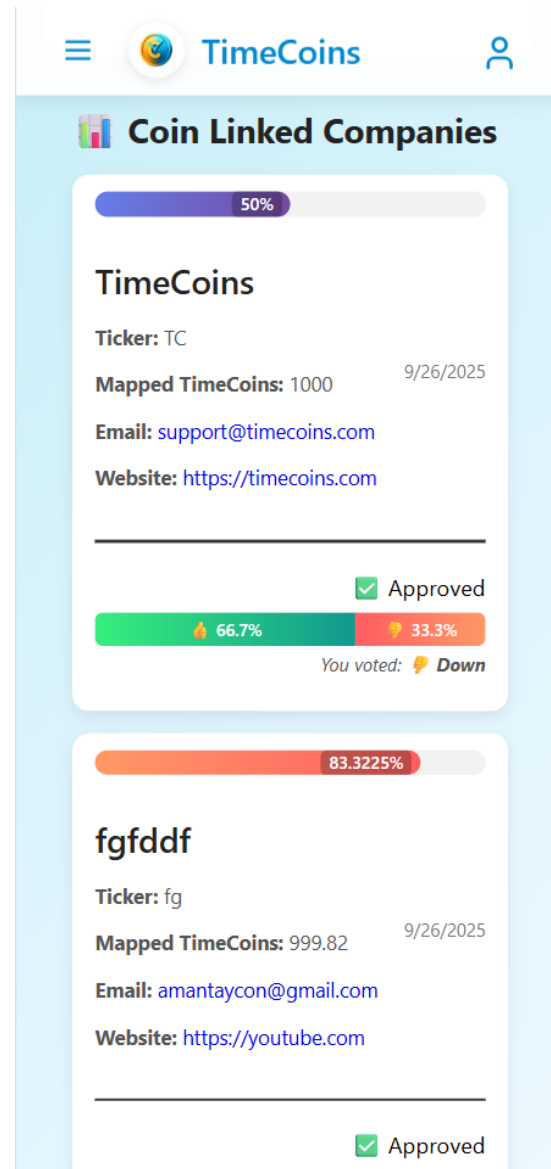


Figure 7.7: Coin Linked Companies Page

7.2.8 Admin Panel – Manage Coin Linked Companies

This is the control center of **TimeCoins**, where admins can **add new companies**, **monitor community votes**, and **approve or reject** them based on trust and transparency. Every approved company directly impacts the **TimeCoins market value**, ensuring a decentralized and fair ecosystem.

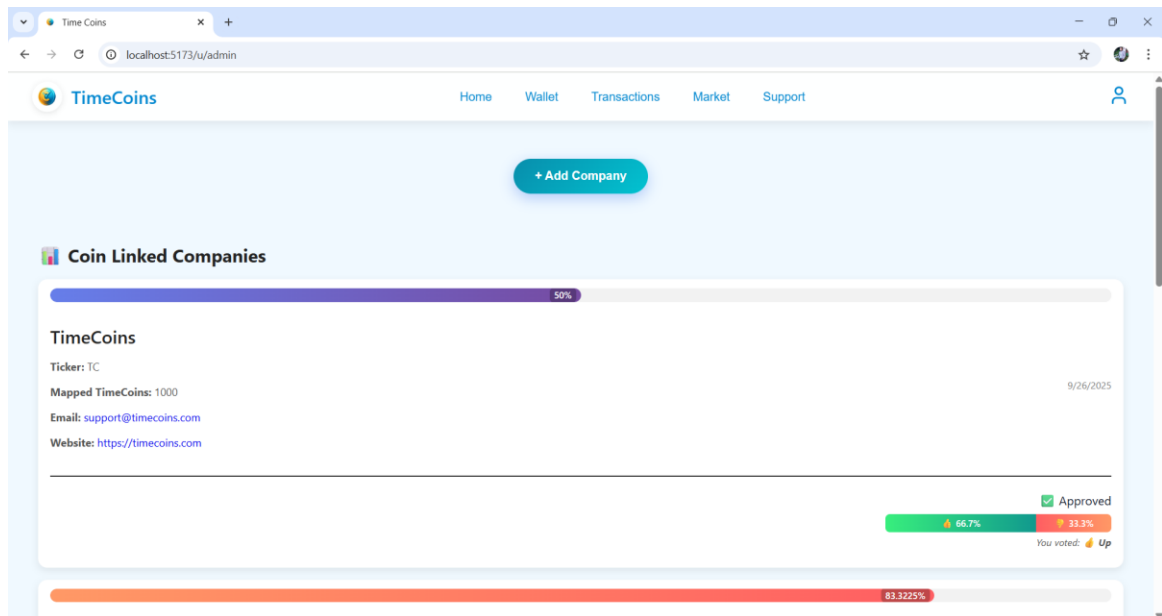


Figure 7.8: Behind every secure coin lies a transparent admin system. Managing companies, tracking votes, and maintaining trust — all in one powerful panel.

7.2.9 Profile Page – Personalized User Dashboard

The Profile Page in the **TimeCoins System** allows each user to view and manage their personal details in a clean and structured layout. It displays essential information such as the user's **profile picture**, **full name**, **username**, **bio**, and **role**. Additionally, it includes a quick summary of the user's activity—showing **total transactions** and **membership date**, offering transparency and insight into their engagement with the platform.

The **About section** highlights a brief user description and professional role, while the **Stats section** provides summarized metrics that reflect the user's interaction and credibility within the TimeCoins ecosystem. This personalized view enhances user trust and helps build a sense of digital identity within the platform.

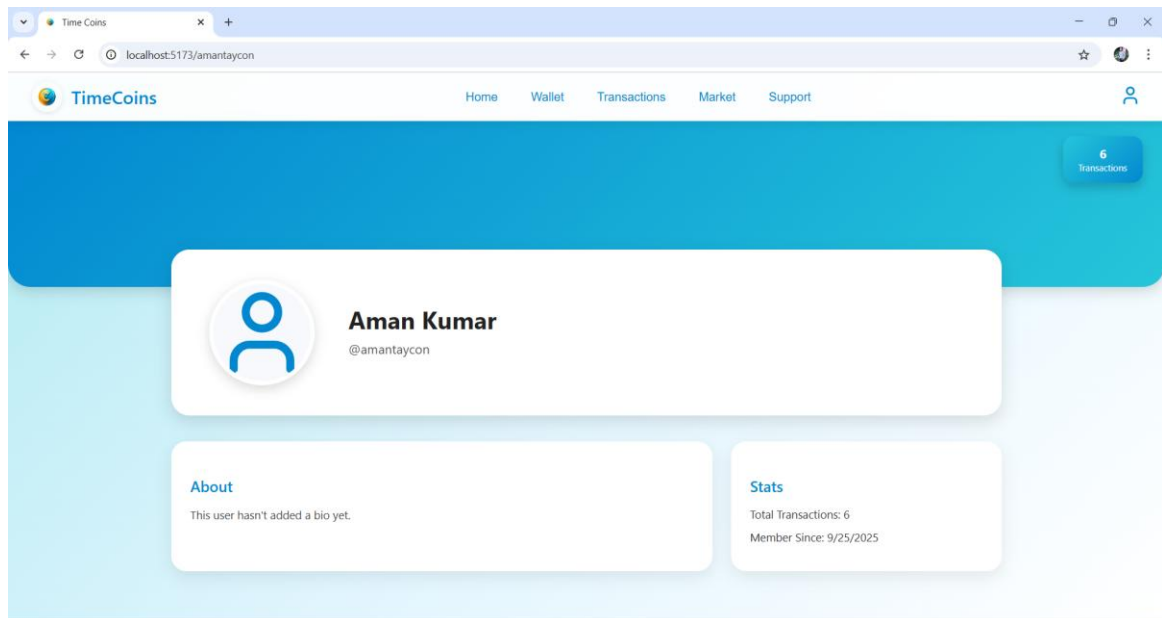


Figure 7.9: Profile page shows the user details, bio, and total number of transactions.

7.2.10 Settings Page

The **Settings Page** allows users to manage their personal information, application preferences, and security options.

Users can update their **full name**, add or modify their **bio**, and view their **username** and **email address** (read-only for security reasons).

Under the **Preferences section**, users can control their **notification settings**, **TimeCoin update alerts**, and toggle between **light or dark mode** for a personalized experience.

In the **Security section**, users can securely change their password by requesting a **reset link** sent directly to their registered email.

After modifying any setting, the user can click the “**Save Changes**” button to update their preferences in real time.

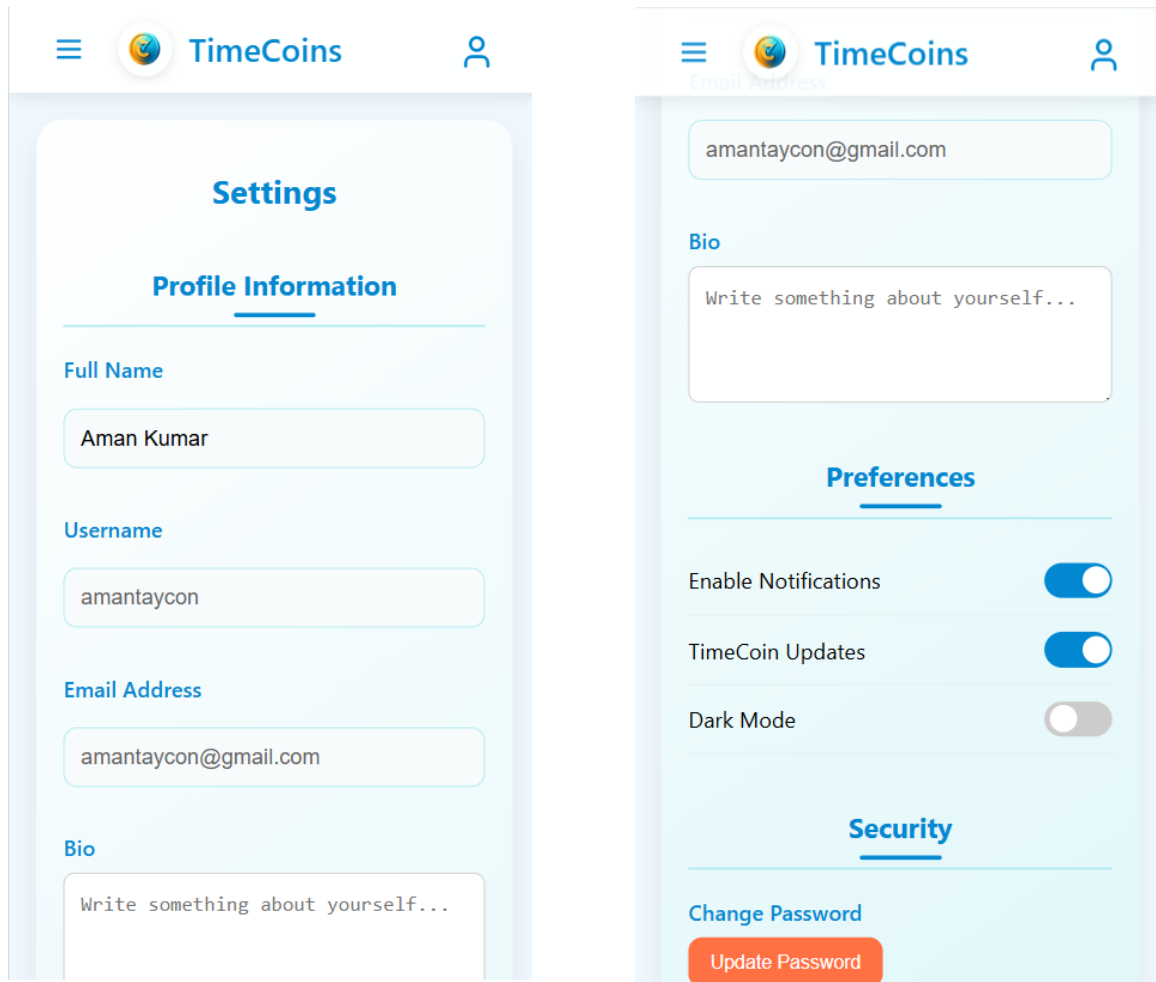


Figure 7.10: Settings page allows users to update personal details, manage notifications, enable dark mode, and reset passwords securely.

7.3 Result Analysis

The developed system meets all primary objectives, offering:

- **Secure transactions** through Spring Boot and JWT.
- **Efficient database design** with proper normalization and indexing in MySQL.
- **Dynamic TimeCoin valuation** based on company share performance.
- **Smooth real-time communication** using WebSocket integration.
- **Responsive and user-friendly interface** for both users and admins.

Testing showed the system performed efficiently under multiple concurrent operations with no major data inconsistency.

7.4 Conclusion

The **TimeCoins** project successfully demonstrates how digital coins can represent real-world company shares and be used in everyday transactions.

It bridges financial systems and digital communication, enabling users to manage wealth and interact in a unified ecosystem.

This innovation opens a new era where products can be purchased indirectly using company-backed digital coins.

Future enhancements may include:

- Mobile application development for accessibility.
- Advanced analytics dashboards for share and coin prediction trends.

CHAPTER 8

Conclusion and Future Enhancements

8.1 Conclusion

The *TimeCoins* project successfully demonstrates a modern approach to digital currency by linking its value directly to company shares. This innovative model not only stabilizes the coin's worth but also encourages transparent and investment-driven digital transactions. Through its structured voting mechanism, users play an essential role in approving companies, ensuring community participation and trust.

The system is built with a secure backend using **Spring Boot**, a dynamic frontend using **Vite + React**, and **MySQL** as the database, ensuring reliability, scalability, and efficiency. Real-time communication through **WebSocket hybrid mode** enhances user experience by allowing instant message updates while maintaining HTTP-based requests for message delivery.

Overall, *TimeCoins* successfully integrates digital currency management, company share mapping, and secure communication into one unified platform. It represents a meaningful step toward merging financial technology with blockchain-inspired value systems.

8.2 Future Enhancements

While *TimeCoins* achieves its core objectives, several enhancements can be implemented in the future to expand its capabilities:

- 1. Mobile Application Integration:**

Developing Android and iOS applications to improve accessibility and reach a wider user base.

- 2. AI-Based Value Prediction:**

Implementing machine learning models to predict coin value fluctuations based on company performance and market trends.

- 3. Advanced Analytics Dashboard:**

Adding visual dashboards for users and administrators to monitor coin circulation, transaction history, and company performance metrics.

- 4. Enhanced Security Features:**

Including two-factor authentication (2FA), biometric logins, and encryption upgrades for improved account safety.

5. E-Commerce Integration:

Partnering with online marketplaces (e.g., Flipkart-like platforms) to allow users to buy products directly using TimeCoins, effectively enabling share-based shopping.

References

The references listed below include the books, research papers, websites, and online resources that were used to understand the technologies and concepts implemented in the TimeCoins project.

Books

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2. Craig Walls, *Spring in Action*, Manning Publications.
3. Thomas A. Powell, *HTML & CSS: The Complete Reference*, McGraw Hill.
4. Kyle Simpson, *You Don't Know JS*, O'Reilly Media.
5. Robin Nixon, *Learning PHP, MySQL & JavaScript*, O'Reilly Media.

Research Papers

1. Satoshi Nakamoto, *Bitcoin: A Peer-to-Peer Electronic Cash System*, 2008.
2. Narayanan, Arvind, et al., *Bitcoin and Cryptocurrency Technologies*, Princeton University Press, 2016.

Web Resources

1. <https://spring.io> – Official Spring Boot Documentation.
2. <https://react.dev> – Official React Documentation (for UI references).
3. <https://www.mysql.com> – MySQL Database Documentation.
4. <https://developer.mozilla.org> – MDN Web Docs for JavaScript, HTML, and CSS.
5. <https://www.coinmarketcap.com> – For studying coin valuation and crypto market references.
6. <https://www.geeksforgeeks.org> – For understanding backend and API integration concepts.
7. <https://stackoverflow.com> – For troubleshooting and community-based coding solutions.